# The Digi-Comp II: an illustrated manual

*West bay 3: how the Digi-Comp II computes, operation by operation · 27 October 2026*

How Babbage's Hall's Digi-Comp II computes, operation by operation: count, clear, add, multiply, complement, subtract and divide, and what follows from them (AUTO is still to be built), with the parts each one needs, for anyone who wants to build only some of it.

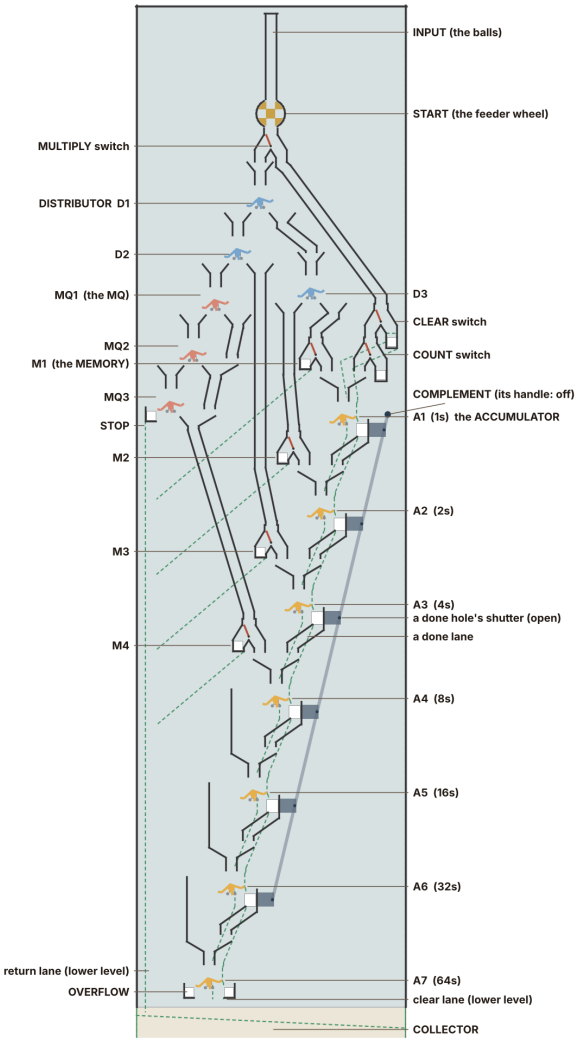
Every picture of the board is drawn from `digicomp2-geometry.lua`, the same numbers the simulation and the printed parts use (`manual/make-figures.sh` redraws them); only the arrows and labels are added. Where this board does a thing differently from E.S.R.'s, the original's way is given too, from its manual (Evil Mad Scientist's copy: cdn.evilmadscientist.com/KitInstrux/DCII-manual.pdf).

**The whole idea in one paragraph.** A flip-flop is a rocker that a ball always tips over, and the ball leaves on the side the rocker used to read. Chain them so that a ball leaving one side goes on to the next, and a ball adds one in binary, carrying as it goes: that is the accumulator. Let a set of switches (the memory) choose where balls go into it, and they add a number. Let three flip-flops (the distributor) deal the balls out to those switches in turn, and every four balls add the memory once. Let three more (the MQ) count those turns down, and stop when they run out, and the machine multiplies. Cover the holes where balls finish, so a ball goes on through every flip-flop and turns each one over, and the machine finds a number's complement, which subtracts. Everything else on the board is lanes and holes.

**Contents**

1. The parts: the flip-flop, the switch, the registers, reading binary
2. Counting
3. Clearing
4. Adding
5. Multiplying, and what else that does
6. The complement: subtracting, counting out, dividing; AUTO (still to come)
7. Which parts each operation needs, and building only part of it
8. Settings at a glance

# 1. The parts

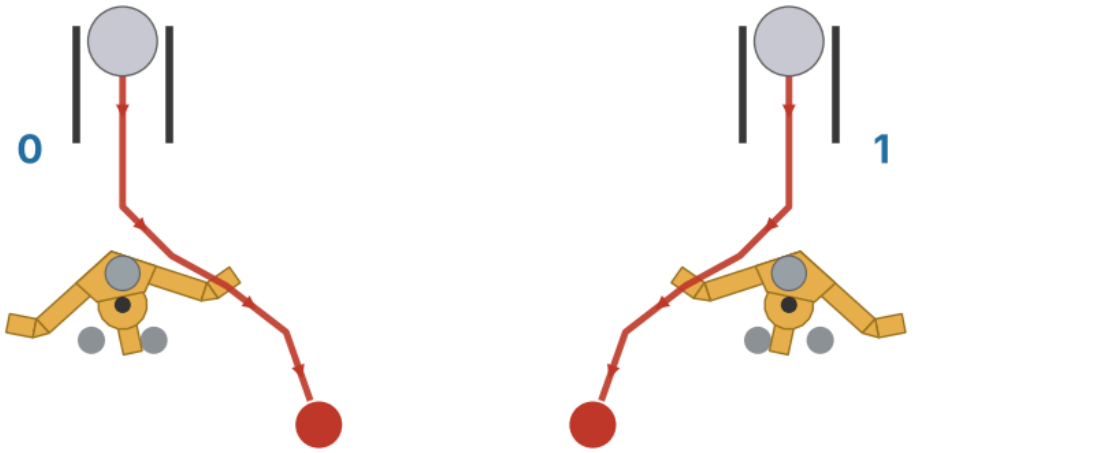



*The board, seen from above. The upper plate (blue-grey) carries the balls while they compute; the lower level (dashed green walls) carries finished balls to the collector.*

The board slopes toward you at 15 degrees, so a ball let go at the top rolls down it. It has two levels. The balls compute on the **upper plate**. A ball that has finished its work drops through a hole to the **lower level** and rolls under everything else to the **collector** at the bottom. Every ball ends there.

There are only two kinds of moving part, the flip-flop and the switch.

## The flip-flop




*A flip-flop: found at 0, the ball leaves right and the rocker turns to 1; found at 1, it leaves left and the rocker turns to 0.*

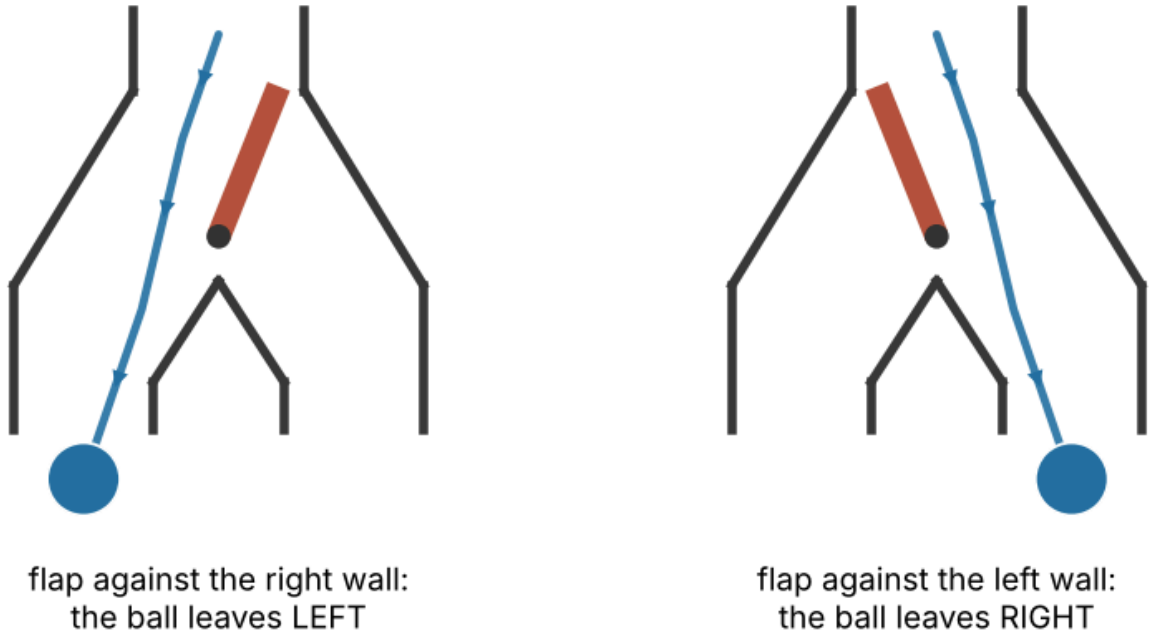
A **flip-flop** is a rocker on a pin: a roof with a cup at each end, a small steel ball pressed in under its apex for weight, and a tail that rests on one of two stop pegs. Its weight sits a little above the pin, so it falls over to one side or the other and stays there. That is one bit of memory: **apex right reads 1, apex left reads 0** (the 0 and 1 are printed on the plate on either side of each apex).

A ball rolling down onto it meets the roof on its raised side, rolls into that side's cup, and its weight tips the rocker over; at the end of the swing the cup tips the ball out. So **every ball flips the rocker**, and **leaves on the side the rocker used to read**:

| The rocker read | It now reads | The ball leaves |
|---|---|---|
| 0 | 1 | to the right |
| 1 | 0 | to the left |

That is all a flip-flop does. What it is used for depends on where its two exits lead. All thirteen on this board are the same printed part.

## The switch




*A switch: the flap closes one outlet; the ball takes the other.*

A **switch** is a flap on a pin between two outlets, set by hand. Its free end rests against one wall of the inlet and closes the outlet on that side. A ball only pushes the flap harder against the wall and leaves by the other outlet, so the switch stays as it was set. Each switch has an "on" side, printed on the plate.

## The registers and switches

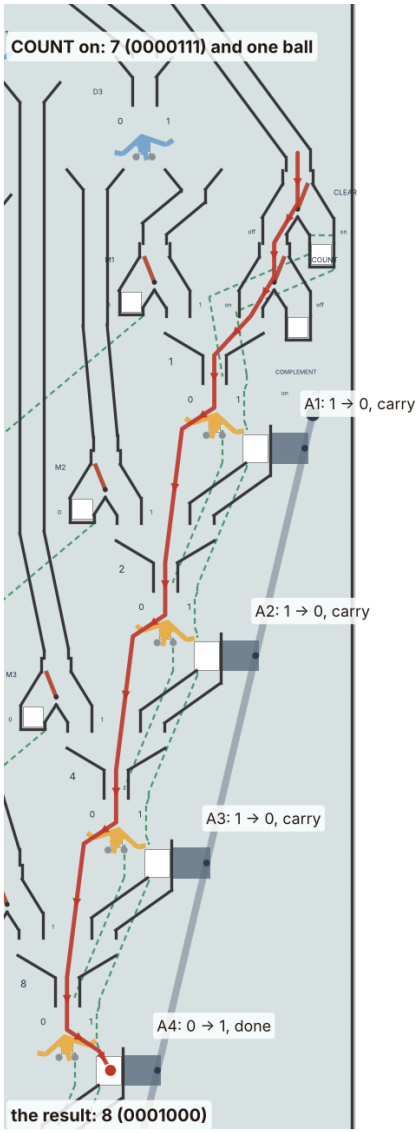
| Name | What it is | What it does |
|---|---|---|
| **START** | the feeder: a wheel with four notches under the **INPUT** channel of waiting balls | each quarter turn of its knob lets one ball out |
| **MULTIPLY** | a switch | on: the ball goes to the distributor. Off: down to CLEAR |
| **CLEAR** | a switch | on: the ball drops into the clear lane. Off: on to COUNT |
| **COUNT** | a switch | on: the ball goes into A1. Off: it drops out |
| **A**, the accumulator | seven flip-flops, A1 (1s) to A7 (64s) | holds the answer, 0 to 127 |
| **M**, the memory | four switches, M1 (1s) to M4 (8s) | holds a number, 0 to 15, set by hand |
| **MQ**, multiplier-quotient | three flip-flops, MQ1 (1s) to MQ3 (4s) | counts down by one each cycle; 0 to 7 |
| **D**, the distributor | three flip-flops, D1 to D3 | deals the balls out four ways, in turn |
| **STOP** | a hole beside MQ3 | a ball dropping through it means "finished" |
| **OVERFLOW** | a hole beside A7 | a carry out of A7 (past 127) drops through it |
| **COMPLEMENT** | six shutters over A1 to A6's done holes, on one bar with a handle | on: a ball goes on through every flip-flop below where it comes in, turning each over (section 6) |

### Reading binary

Each flip-flop of A stands for a power of two: A1 the 1s, A2 the 2s, A3 the 4s, and on to A7, the 64s (the numbers are printed beside them). A number is the sum of the places that read 1. Written the usual way, the 64s first: **0101101** is 32 + 8 + 4 + 1 = **45**. The sign beside the machine shows each register in binary and as a number, but the rockers are the machine's own reading.

## 2. Counting

**Set up:** A as you want it (0 to start), MULTIPLY off, CLEAR off, COUNT on. **Then:** START, once for each ball.




*Counting: 7 (0000111) and one ball. A1, A2 and A3 each turn 1 to 0 and pass the ball on (the carry); A4 turns 0 to 1 and the ball drops out through A4's hole. The result is 8 (0001000).*

The accumulator's flip-flops form a staircase. Each one's **right** exit is a hole to the lower level, so the ball is finished. Its **left** exit runs down into the next flip-flop. So a ball arriving at a flip-flop:

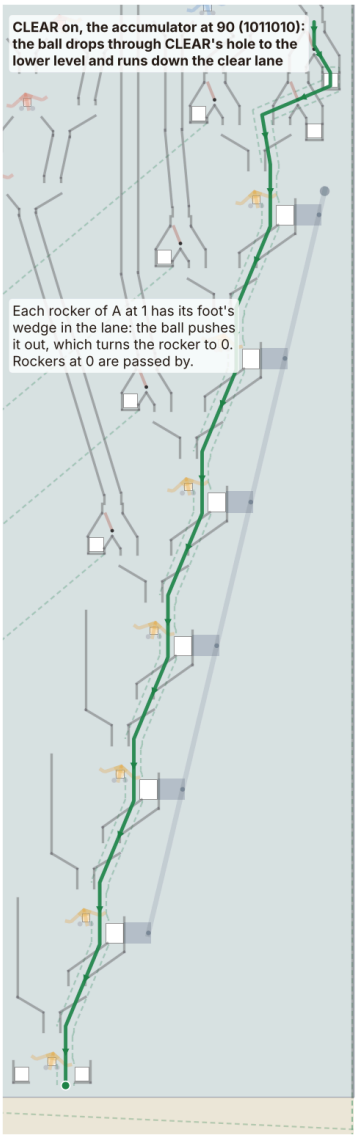
- finds it at **0**: sets it to 1 and drops out. One has been added.
- finds it at **1**: sets it to 0 and goes on to the next flip-flop, which is the **carry** (1 + 1 = 10 in binary).

A ball let into A1 therefore **adds one**, carrying as far as it must. A ball let into A3 adds four, and so on: the memory uses this when it adds.

**Overflow.** At 127 (1111111) the next ball carries through all seven and out of A7 through the OVERFLOW hole, leaving 0000000.

## 3. Clearing

**Set up:** MULTIPLY off, CLEAR on. **Then:** START, one ball.




*CLEAR from 90: the ball drops through CLEAR's hole and runs down the clear lane on the lower level, past the foot of each of A's flip-flops.*

Under each flip-flop of A, on the lower level, is its **foot**: an arm joined to the rocker through the plate, ending in a wedge. When the rocker reads 1, its wedge leans into the **clear lane**. A ball sent down the clear lane pushes every wedge it meets back out of its way, and pushing the wedge turns the rocker to 0. Rockers already at 0 are passed by. **One ball clears the whole accumulator.**

(You can also clear by hand: set every rocker of A to 0.)

## 4. Adding

**Set up:** A = the first number; M = the second (at most 15); MQ = 1; the distributor set to start (D1 = 1, D2 = 1, D3 = 0; key `D`); MULTIPLY on. **Then:** START until a ball drops through STOP: **five balls.**

Adding is a multiply by 1 (section 5): the first four balls add M to A, and the fifth finds the MQ used up and stops. Three things make it work.

**The memory.** Each memory switch stands beside one of A1 to A4. Set to 1, it sends a ball that reaches it into that flip-flop, which adds 1, 2, 4 or 8 there (with any carry). Set to 0, it drops the ball out. The memory itself never changes.

**The distributor** deals the balls out in turn. D1 sends them to D2 and D3 alternately, and each of those alternately to its two sides: four balls, four places, then round again. Set to start, the four balls of each cycle go:

| Ball of the cycle | Way through the distributor | Then |
|---|---|---|
| 1st | D1 left, D2 left | through the MQ (one taken away), on to M4 |
| 2nd | D1 right, D3 right | M1 |
| 3rd | D1 left, D2 right | M3 |
| 4th | D1 right, D3 left | M2 |

So **every four balls add the memory to A once.**

**The MQ** counts the cycles (section 5).

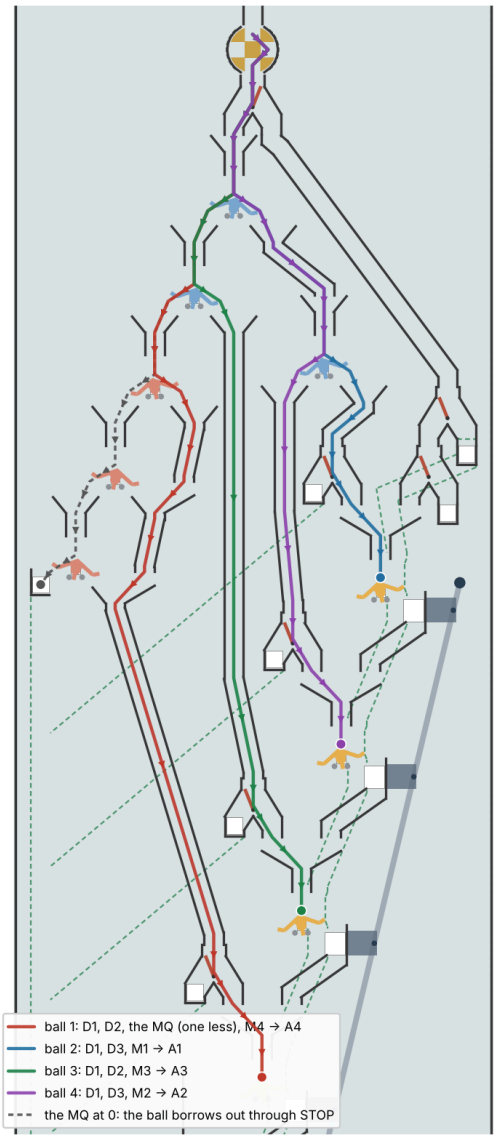
**Worked example: 37 + 11** (the tour's). A = 0100101 (37), M = 1011 (11: M4, M2, M1 on; M3 off), MQ = 1:

| Ball | Goes to | A after it | MQ after it |
|---|---|---|---|
| 1 | the MQ (1 to 0), then M4 = 1: adds 8 | 0101101 = 45 | 0 |
| 2 | M1 = 1: adds 1 | 0101110 = 46 | 0 |
| 3 | M3 = 0: drops out | 0101110 = 46 | 0 |
| 4 | M2 = 1: adds 2 (carries through A2, A3, A4 to A5) | 0110000 = 48 | 0 |
| 5 | the MQ, at 0: through STOP | **0110000 = 48** | 7 |

Keep the sum at 127 or less (the total passes through OVERFLOW otherwise).

## 5. Multiplying

**Set up:** A = 0 (CLEAR it); M = the multiplicand (at most 15); MQ = the multiplier (at most 7); distributor set to start; MULTIPLY on. **Then:** START until a ball drops through STOP: **4 x MQ + 1 balls.**




*One cycle of four balls with MULTIPLY on (D1 = 1, D2 = 1, D3 = 0 at the start), and, dashed, the ball that finds the MQ at 0 and leaves through STOP.*

Multiplying is adding over and over: every four balls add M to A, and the MQ counts how many times.

**The MQ** is three flip-flops in a staircase like A's, but it is read the other way round: **1 with the apex to the left** (printed so on the plate). Read that way, a ball through it **takes one away**. A ball that finds an MQ flip-flop at 1 sets it to 0 and leaves to the right, which means done; it runs on to M4. One that finds it at 0 sets it to 1 and goes on to the next, which is the **borrow** (10 - 1 = 01). A borrow out of MQ3 means the MQ was already 0. That ball drops through **STOP**, and the hand (you) stops turning the knob.

**Worked example: 13 x 3** (the tour's). M = 1101 (13: M4, M3, M1 on), MQ = 3, A = 0:

| Balls | Cycle | A after the cycle | MQ |
|---|---|---|---|
| 1 to 4 | first: adds 8, 1, 4, 0 | 13 | 2 |
| 5 to 8 | second | 26 | 1 |
| 9 to 12 | third | 39 | 0 |
| 13 | the MQ at 0: STOP | **0100111 = 39** | 7 |

Put the smaller number in the MQ: 5 x 7 takes 21 balls with 5 in the MQ, 29 with 7.

### What else multiplying does

These need nothing more than the multiply set-up, with different numbers.

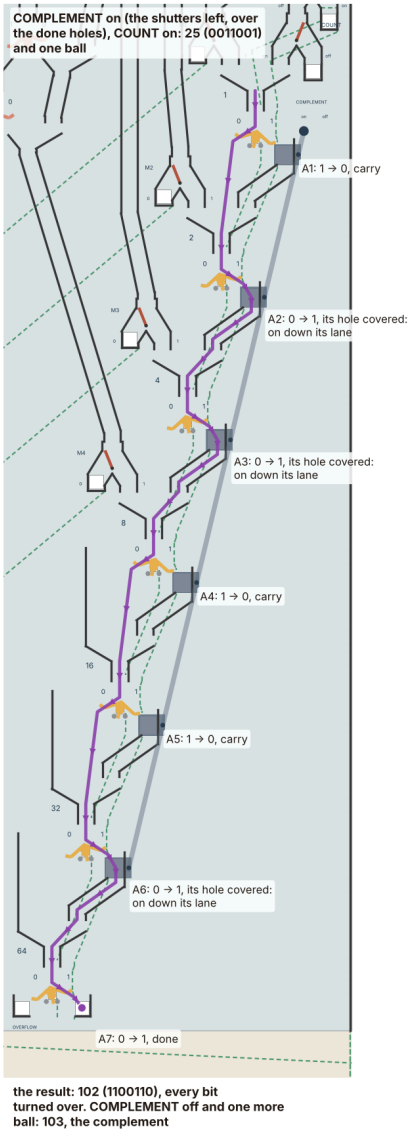
- **Multiply and add.** A isn't cleared by MULTIPLY, so whatever is in A is added to: A + M x MQ. Two multiplies in a row add up: 4 x 14, then 5 x 13 without clearing, gives 56 + 65 = **121**.
- **Squaring.** The same number in M and the MQ: 7 x 7 = **49** (29 balls).
- **Shifting.** Multiplying by 2, 4 or 8 moves a number's bits one, two or three places left: put 2, 4 or 8 in M and the number in the MQ, or the number in M and 2 or 4 in the MQ. M = 1101 (13), MQ = 4: **0110100** (52).
- **Decimal to binary** (from E.S.R.'s manual). To find 57 in binary, write it as 5 x 10 + 7: A = 7, M = 10 (1010), MQ = 5. Twenty-one balls later A reads **0111001**. (The MQ's 7 limits this to 79; split bigger numbers, as the original manual does for 123 = 64 + 59.)

## 6. The complement: subtracting, counting out, dividing

### The complement

Subtracting is done by adding a **complement**. The two's complement of a 7-bit number is what you add to it to make 128 (0000000 with an overflow). Turn every bit over (the **ones' complement**), then add one:

| | binary | |
|---|---|---|
| 25 | 0011001 | |
| every bit turned over | 1100110 | the ones' complement: 102, which is 127 - 25 |
| plus one | **1100111** | the complement: 103, which is 128 - 25 |




*COMPLEMENT on: one ball goes through all seven flip-flops of A and turns each one over. Where it sets one to 1, the done hole is covered, so it rolls on down that one's done lane into the next.*

**On this board.** Each of A1 to A6 has, below its done hole, a **done lane** that runs down and across into the next flip-flop's funnel. The hole is open as a rule, and a ball that sets the flip-flop to 1 drops through it, done; the lane below is never reached. The **COMPLEMENT** switch is six **shutters**, one over each of those holes, all on one bar (its handle is beside A1): slid to the left (on), they cover the holes. Now a ball that sets a flip-flop to 1 rolls over the covered hole and down the done lane into the next flip-flop, just as one that sets it to 0 goes on to the next by the carry. Either way it goes on, so it turns over every flip-flop from where it came in to A7. (Out of A7 it drops through A7's done hole, which has no shutter, or through OVERFLOW.)

1. A = the number (count it in, or set it). COUNT on, MULTIPLY and CLEAR off.
2. COMPLEMENT on, START: **one ball**. A now holds 127 - A.
3. COMPLEMENT off, START: **one ball** adds one. A holds 128 - A, the complement.

**On the original** it is one START with COMPLEMENT on, and the machine does four balls by itself (in AUTO): six **mode** flip-flops T1 to T6 between A's flip-flops, and two more, CF1 and CF2, to sequence them. Ball 1 sets T1 to T6 to COMPLEMENT mode, ball 2 passes every bit of A and turns each over, ball 3

sets them back to ADD mode, ball 4 adds one. This board has no AUTO yet, so the hand does what CF1 and CF2 do: it turns COMPLEMENT off between the two balls. The shutters do the work of T1 to T6.

**Negative numbers.** Read this way, a number whose leftmost bit is 1 is negative: 1100111 is -25. Seven bits hold -64 to +63.

## Subtracting

A = the number to take away. Complement it (two balls, as above). Then MULTIPLY on, M = the number to take it from, MQ = 1, D set to start, START until STOP (five balls: an add). With COMPLEMENT off the adding is ordinary.

**Example: 12 - 4** (in the tour). A = 4 (0000100); complemented: 1111100 (124). Add M = 12: 124 + 12 = 136, which goes over 128: the ball that carries out of A7 drops through OVERFLOW, and A is left with **0001000 = 8**. The overflow is simply lost, which is what makes the complement subtract.

## Counting out a number of balls

A = N; complement it (two balls); then COUNT on, MULTIPLY off. A counts up from 128 - N, and the **N-th ball** carries out of A7 through OVERFLOW. Stop there: that is N balls. (On the original, OVERFLOW set to HALT stops the machine in AUTO; here the hand watches the OVERFLOW hole. The tour doesn't run this one: it is counting, which the tour checks, from a complement.)

## Dividing

Dividing is subtracting over and over, counting how many times: adding the divisor to the dividend's complement until the sum goes over 128, while the MQ counts the cycles.

1. A = the dividend; complement it (two balls).
2. M = the divisor, MQ = 7 (111), D set to start, MULTIPLY on.
3. START, ball after ball. Each cycle of four balls adds M to A and takes one from the MQ (its first ball). When a ball drops through **OVERFLOW**, go on to **the end of that cycle**: stop when the next ball would go to the MQ (D1 and D2 both read 1, as D sets them to start).
4. Read the MQ's **lower numbers** (the other reading of its rockers: lower = 7 - upper; upper 100 is lower 011): that many cycles were begun. - **A reads 0**: the divisor went exactly. The quotient is the cycles, the remainder 0. - **Otherwise** the last cycle went over: the quotient is one less than the cycles, and the **remainder is M - A**.

**Example: 13 / 5** (in the tour). 13 complemented is 115. The cycles add 5: 120, 125, and in the third 130, which goes over (the ball that adds 4 drops through OVERFLOW) and leaves 2. The cycle ends; the MQ reads upper 100, lower 011 = 3 cycles. A is 2, not 0: quotient 3 - 1 = **2**, remainder 5 - 2 = **3**. (13 = 2 x 5 + 3.)

**Example: 25 / 5.** 25 complemented is 103; 108, 113, 118, 123, and in the fifth cycle 128: over, leaving 0. Five cycles, A = 0: quotient **5**, remainder 0.

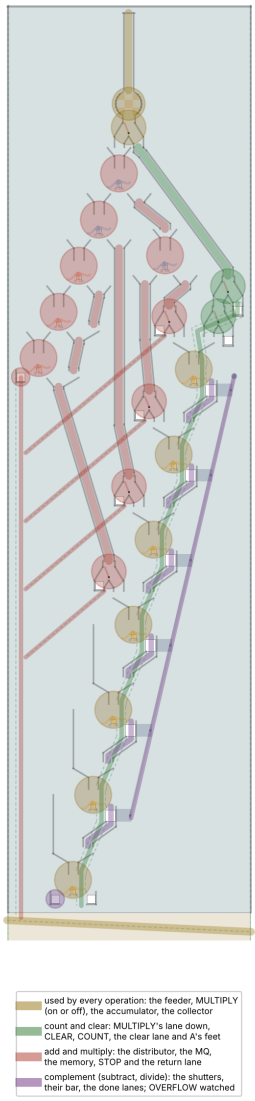
**Why to the end of the cycle.** A cycle adds M one bit at a time (on this board its 8s, then 1s, 4s, 2s), and A may go over 128 on any of them. Stopping at that ball would leave the rest of M unadded, and the remainder wrong; finished, the cycle has added all of M, and A = cycles x M - dividend. (E.S.R.'s manual stops at the overflow, in AUTO, and has a chapter on remainders that could not be read here; this rule is this board's own, and it is what the tour checks.)

**Limits.** With the MQ at 7, at most seven cycles: quotients up to 6 (7 when exact). If the MQ runs out first, a ball drops through STOP: the quotient was too big for it.

### Running by itself: AUTO

On the original, AUTO lets a ball reaching the bottom roll under the START lever and release the next ball, so a whole multiply runs by itself. The ball through STOP takes a way that misses the lever, so the machine stops. Here every ball is let in by turning the knob, which is the original's MANUAL. For AUTO, this machine will let a wound weight turn the wheel, held by an anchor, as a clock's weight turns its wheels and its escapement holds them: each push on the anchor's arm lets the wheel round a quarter turn, one ball. That part works on its own (`tests/escapement.lua`); the ball at the bottom pushing the arm, and STOP's ball not pushing it, are still to be built (`FINDINGS.md` has the tries so far).

## 7. Which parts each operation needs




*Which parts of this board each kind of operation uses.*

| Operation | Feeder | A | CLEAR, clear lane | COUNT | MULTIPLY | D, MQ, STOP | M | COMPLEMENT (shutters, done lanes) | OVERFLOW (watched) |
|---|---|---|---|---|---|---|---|---|---|
| Count | yes | yes | | yes | (off) | | | | |
| Clear | yes | yes | yes | | (off) | | | | |

| Operation | Feeder | A | CLEAR, clear lane | COUNT | MULTIPLY | D, MQ, STOP | M | COMPLEMENT (shutters, done lanes) | OVERFLOW (watched) |
|---|---|---|---|---|---|---|---|---|---|
| Add | yes | yes | | | yes | yes | yes | | |
| Multiply, multiply-add, square, shift | yes | yes | | | yes | yes | yes | | |
| Decimal to binary | yes | yes | | | yes | yes | yes | | |
| Complement | yes | yes | | yes | (off) | | | yes | |
| Subtract | yes | yes | | yes | yes | yes | yes | yes | |
| Count out N balls | yes | yes | | yes | (off) | | | yes | yes |
| Divide | yes | yes | | yes | yes | yes | yes | yes | yes |

(MULTIPLY "(off)": on this board every ball passes the MULTIPLY switch on its way down; a board without it would feed CLEAR directly.)

### Building only part of it

The steps this machine was built in are each a complete, smaller machine, kept in the repository's history. Each has its own board, tiles, parts list and `make-stls.sh`:

| To build | Take | It does | Board |
|---|---|---|---|
| A three-bit counter, one level, balls into bins | step 1 (commit `cb8bb07`) | count to 7 | the smallest |
| The accumulator with START, CLEAR and COUNT, two levels | step 2a (commit `201ee23`) | count to 127, clear | 322 × 878 mm, 10 + 10 tiles |
| Count, clear, add, multiply | step 2b (commit `71d155a`) | and section 5's uses | 424 × 1216 mm, 14 + 14 tiles |
| Everything so far | this folder (step 2c) | all of the above, and complement, subtract, divide | 424 × 1573 mm, 18 + 18 tiles |

To get an earlier step's files without changing your checkout:

```
git archive 201ee23 demo/WyomingWill/BabbagesHall/exhibits/DigiComp2 | tar -x -C /some/other/folder
```

Its README lists the parts to print and buy. Some other ways to build less:

- **Without CLEAR**, clear by hand (set every rocker to 0). The feet are part of every rocker either way; the clear lane's walls would have to be taken out of `digicomp2-geometry.lua` to leave them off the lower tiles.
- **Fewer memory bits or a shorter MQ** limit the numbers (M up to 15, MQ up to 7 now). That would need the layout redrawn: it is set in `digicomp2-geometry.lua` and has only been built and tested as it is.
- **For adding only**, the MQ is still needed: it is what stops the fifth ball.
- **Without COMPLEMENT**, leave out the shutters and their bar: the done holes stay open, the done lanes are never reached, and the board is what step 2b's was, only longer. (A shorter board without the lanes is step 2b's.)

## 8. Settings at a glance

| Operation | A | M | MQ | D | MULTIPLY | CLEAR | COUNT | Balls |
|---|---|---|---|---|---|---|---|---|
| Count | start value | | | | off | off | on | one each |

| Operation | A | M | MQ | D | MULTIPLY | CLEAR | COUNT | Balls |
|---|---|---|---|---|---|---|---|---|
| Clear | | | | | off | on | | 1 |
| Add A + M | A | M | 1 | start | on | | | 5 |
| Multiply M x MQ | 0 | M | MQ | start | on | | | 4 x MQ + 1 |
| Multiply-add A + M x MQ | A | M | MQ | start | on | | | 4 x MQ + 1 |
| Complement A | A | | | | off | off | on | 1 with COMPLEMENT on, 1 with it off |
| Subtract M - A | complement of A | M | 1 | start | on | | | 5 |
| Count out N | complement of N | | | | off | off | on | until OVERFLOW: N |
| Divide A / M | complement of A | M | 7 | start | on | | | until OVERFLOW, then to the end of that cycle |

On this board, by hand (keys in the simulation): A's rockers `1` to `7`, the memory `H J K L`, the MQ `8 9 0`, the distributor to start `D`, MULTIPLY `M`, CLEAR `C`, COUNT `N`, COMPLEMENT `X`, START `Space`. The tour (`T`) runs counting, clearing, 37 + 11, 13 x 3, the complement of 25, 12 - 4 and 13 / 5. Before each ball, in the tour or with `Space`, the exhibit draws the way the ball will go as a yellow dashed line on the board, and its sign says what the ball will do and what the accumulator was and will be: a good way to follow this manual on the machine itself.